AutoInsight — Session Report

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Q1. how many sales did we make in 2024?

We made **300 sales in 2024**. This figure represents the total number of transactions recorded between January 1, 2024 and December 31, 2024.

SQL

```
SELECT COUNT(*) AS total_sales_2024 FROM sales AS s WHERE s.sale_date >= '2024-01-01' AND
s.sale_date < '2025-01-01' LIMIT 1000
```

Rows affected: 1

Q2. monthly total revenue over the last 2 years

There is no revenue data available for the requested period.

SQL

```
SELECT DATE_TRUNC('MONTH', s.sale_date) AS month, SUM(s.sale_price) AS total_revenue FROM
sales AS s WHERE s.sale_date >= (DATE_TRUNC('YEAR', CURRENT_DATE) - INTERVAL '1 YEAR') GROUP
BY month ORDER BY month LIMIT 1000
```

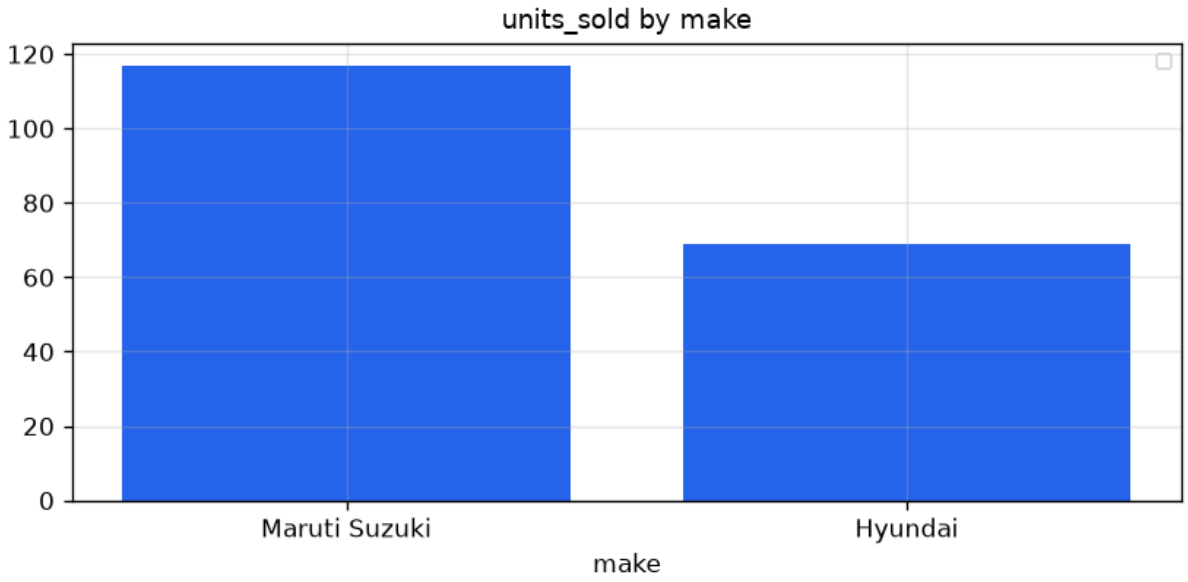
Rows affected: 0

Q3. top 5 car models by units sold

The top five models by units sold are: Maruti Suzuki Baleno with 117 units, Hyundai i20 with 69 units, Maruti Suzuki Swift with 66 units, Maruti Suzuki Alto K10 with 55 units, and Hyundai Grand i10 Nios with 45 units. This shows Maruti Suzuki leading the sales chart, followed by Hyundai.

SQL

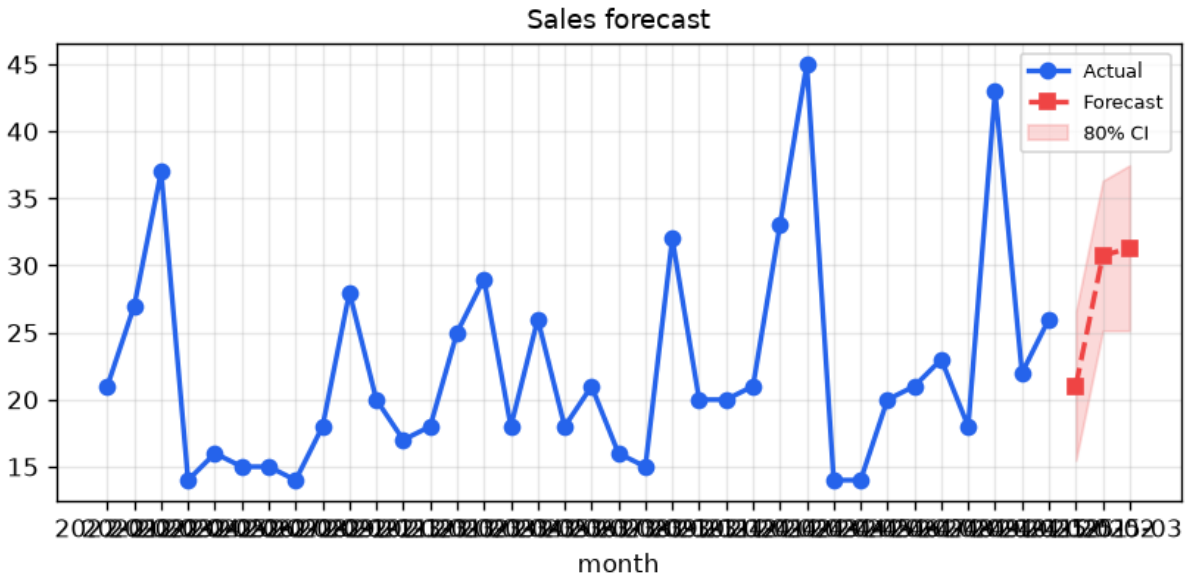
```
SELECT v.make, v.model, COUNT(*) AS units_sold FROM sales AS s JOIN vehicles AS v ON
s.vehicle_id = v.id GROUP BY v.make, v.model ORDER BY units_sold DESC LIMIT 5
```



Rows affected: 5

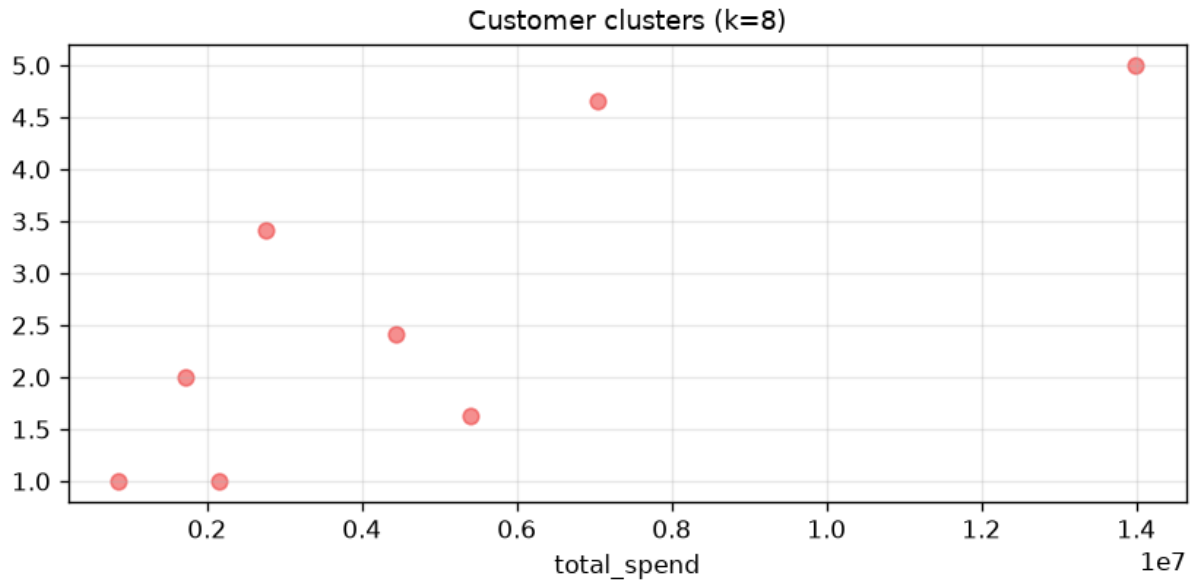
Q4. forecast sales for the next 3 months

The forecast shows an overall upward trend, with sales expected to rise from about 21 units in January to roughly 31 units by March. A noticeable seasonal spike appears in February, where the predicted sales jump to around 30.7 units. For the first month ahead, the model predicts 21.0 units, with an 80 % confidence interval ranging from 15.5 to 26.6 units.



Q5. cluster our customers by spending behavior

The clustering revealed eight distinct spending profiles among 408 customers, with a silhouette score of 0.55 indicating a reasonable separation. The largest group (Cluster 0, 142 customers) are budget first■time buyers, averaging a single purchase of ■8.62 lakh. The smallest but highest■spending segment (Cluster 7, 4 customers) are premium repeat buyers, each spending an average of ■13.99 crore on five purchases. Other clusters fall between these extremes, ranging from moderate■value repeat buyers (Clusters 2 and 4) to high■spend occasional buyers (Clusters 1, 5, 6).



Q6. find anomalies in sales transactions

The Isolation Forest flagged **40 out of 800** sales transactions (5 % contamination) as anomalous. The top offenders exhibit extreme pricing anomalies: sale 487 shows a ■4,886,058.25 sale price against a ■5,192,410.68 listed price with a 6 % discount; sale 669 records a ■1.00 sale price with a ■1,053,361.99 discount (100 % discount_pct); sale 454 mirrors sale 487's high price■to■listed■price ratio; sale 297 shows a ■48,331.91 sale price with a 95 % discount against a ■966,638.28 listed price; and sale 305 has a ■300,000 sale price with an 84 % discount against a ■1,837,489.53 listed price. These extreme discounts and low sale prices relative to the listed price warrant a detailed review by the finance team.

